

ELEVATE LABS TASK

12

Log Monitoring & Analysis

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Domain: Cybersecurity

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1. Understand log types

Logs are automatically generated records that capture events and activities occurring within a system, application, or network. They act as a digital trail, helping administrators and security analysts understand what happened, when it happened, and where it happened. Logs are essential for monitoring system health, troubleshooting issues, detecting security incidents, and performing forensic investigations.

Common Log Types:

- **System Logs**
Record operating system activities such as boot events, hardware issues, and system errors.
- **Application Logs**
Capture events generated by applications, including crashes, user actions, and application-specific errors.
- **Security Logs**
Track security-related events like login attempts, authentication failures, access control changes, and policy violations.
- **Network Logs**
Store information about network traffic, connections, firewall actions, and intrusion detection alerts.
- **Audit Logs**
Maintain a detailed record of user actions and changes made to systems or data, mainly for compliance and accountability.

- **Access Logs**

Log requests made to servers or services (e.g., web server access logs showing IP address, time, and requested resource).

2. Analyze authentication logs

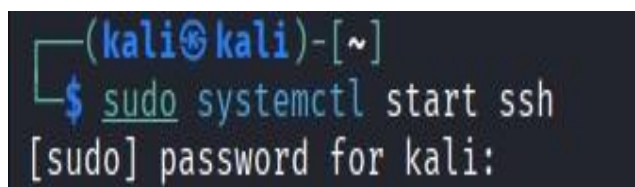
Authentication logs record all events related to user login and logout activities. These logs help identify who accessed the system, from where, when, and whether the attempt was successful or failed. Analyzing authentication logs is critical for detecting unauthorized access, bruteforce attacks, compromised accounts, and insider threats.

Why authentication log analysis is important:

- Detect repeated failed login attempts (brute force)
- Identify logins from unusual IP addresses or locations
- Track privileged (admin/root) access
- Support incident response and digital forensics
- Ensure compliance and accountability

Step 1: Start SSH Service

`sudo systemctl start ssh`



```
(kali@kali)-[~]  
$ sudo systemctl start ssh  
[sudo] password for kali:
```

- **Step 2: Generate Authentication Events**

Login to your own system using SSH: `ssh kali@127.0.0.1` Type yes to accept host key

- Enter wrong password once (failed login)
- Enter correct password (successful login)
- Exit session

```
(kali@kali)-[~]
$ ssh kali@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:+sZqb8GRF2v3ln3qmE8YzVS9yrVTW054HSYgjbVBg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.
kali@127.0.0.1's password:
Permission denied, please try again.
kali@127.0.0.1's password:
Linux kali 6.18.5-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.5-1kali1 (2026-01-19) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

```
(kali@kali)-[~]
$ exit
zsh: corrupt history file /home/kali/.zsh_history
Connection to 127.0.0.1 closed.
```

Step 3: View Authentication Logs

View SSH authentication logs: `sudo journalctl -u ssh`

```
(kali@kali)-[~]
$ sudo journalctl -u ssh
Feb 03 10:45:42 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
Feb 03 10:46:46 kali sshd-session[3835]: Connection closed by 127.0.0.1 port 49538 [preauth]
Feb 03 10:47:43 kali sshd-session[4072]: Connection closed by 127.0.0.1 port 33820 [preauth]
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=kali
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL(9):
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'kali')
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

Step 4: Filter Failed Login Attempts `sudo journalctl | grep -i "failed"`

`journalctl | grep -i "failed"`

```
(kali@kali)-[~]
$ sudo journalctl | grep -i "failed"
Nov 26 23:23:37 kali virtualbox-guest-utils[850]: 04:23:37.871824 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so: cannot open
shared object file: No such file or directory
Nov 26 23:23:37 kali kernel: 04:23:37.871824 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:41 kali wireplumber[964]: default: failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:46 kali kernel: 04:23:46.780207 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:46 kali kernel: 04:23:46.796125 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:46 kali kernel: 04:23:46.821116 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:46 kali kernel: 04:23:46.840007 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:46 kali kernel: 04:23:46.852306 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:47 kali kernel: 04:23:47.376312 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:23:50 kali colord[1350]: failed to get edid data: EDID length is too small
Nov 26 23:23:51 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Avahi.service not found.
Nov 26 23:23:51 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Avahi.service not found.
Nov 26 23:23:53 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.bluez.service': Unit dbus-org.bluez.service not found.
Nov 26 23:23:53 kali dbus-daemon[1067]: [session uid=1000 pid=1067 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': Unit evolution-source-registry.servic
e not found.
Nov 26 23:23:53 kali dbus-daemon[1067]: [session uid=1000 pid=1067 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': Unit evolution-source-registry.servic
e not found.
Nov 26 23:25:20 kali virtualbox-guest-utils[631]: 04:25:20.220497 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so: cannot open
shared object file: No such file or directory
Nov 26 23:25:20 kali kernel: 04:25:20.220497 main rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
Nov 26 23:25:23 kali wireplumber[743]: default: failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:28 kali lightdm[707]: pam_systemd(lightdm-greeter:session): failed to release session: Transport endpoint is not connected
Nov 26 23:25:28 kali lightdm[810]: failed to write utmpx: No such file or directory
```

```
Feb 03 10:42:24 kali dbus-daemon[1300]: [session uid=1000 pid=1300 pidfd=5] Activation via systemd failed for unit 'evolution-source-regis
e not found.
Feb 03 10:42:54 kali kernel: 15:42:54.781179 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_S
Feb 03 10:42:54 kali kernel: 15:42:54.781650 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_N
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PA
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
```

Step 5: Filter Successful Logins `sudo journalctl | grep -i "accepted"`

`journalctl | grep -i "accepted"`

```
(kali@kali)-[~]
$ sudo journalctl | grep -i "accepted"
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
```

Step 6: Check Login History

Successful logins: last

```
(kali@kali)-[~]
$ last

kali      pts/1            127.0.0.1        Tue Feb  3 10:49 - 10:54  (00:04)
kali      tty7             :0              Tue Feb  3 10:42 - still logged in
lightdm   tty7             :0              Tue Feb  3 10:42 - 10:42  (00:00)
kali      tty7             :0              Mon Feb  2 09:02 - 10:28  (01:26)
lightdm   tty7             :0              Mon Feb  2 09:01 - 09:02  (00:00)
kali      tty7             :0              Mon Feb  2 08:43 - 08:59  (00:15)
lightdm   tty7             :0              Mon Feb  2 08:43 - 08:43  (00:00)
kali      tty7             :0              Mon Feb  2 07:36 - still logged in
lightdm   tty7             :0              Mon Feb  2 07:35 - 07:36  (00:00)
kali      tty7             :0              Mon Feb  2 07:24 - 07:32  (00:08)
lightdm   tty7             :0              Mon Feb  2 07:23 - 07:24  (00:00)
kali      tty7             :0              Mon Feb  2 07:14 - still logged in
lightdm   tty7             :0              Mon Feb  2 07:14 - 07:14  (00:00)
kali      tty7             :0              Sat Jan 31 23:05 - 00:13  (01:07)
lightdm   tty7             :0              Sat Jan 31 23:05 - 23:05  (00:00)
kali      tty7             :0              Sat Jan 31 19:38 - 20:12  (00:33)
lightdm   tty7             :0              Sat Jan 31 19:38 - 19:38  (00:00)
kali      tty7             :0              Sat Jan 31 12:21 - still logged in
lightdm   tty7             :0              Sat Jan 31 12:21 - 12:21  (00:00)
kali      tty7             :0              Sat Jan 31 10:33 - 12:07  (01:34)
lightdm   tty7             :0              Sat Jan 31 10:33 - 10:33  (00:00)
postgres pts/2            Sat Jan 31 09:29 - 09:29  (00:00)
kali      Sat Jan 31 09:24 - still logged in
postgres Sat Jan 31 09:23 - 09:23  (00:00)
kali      tty7             :0              Sat Jan 31 08:35 - still logged in
lightdm   tty7             :0              Sat Jan 31 08:34 - 08:35  (00:00)
kali      tty7             :0              Sat Jan 31 07:00 - 07:26  (00:17)
```

3. Identify Failed Logins

Failed login attempts occur when a user provides incorrect credentials or tries to access a system without proper authorization. These events are recorded in authentication logs and are crucial indicators of security threats such as brute-force attacks, password guessing, or unauthorized access attempts. By identifying failed logins, system administrators and forensic analysts can detect suspicious behavior early and take preventive actions like blocking IP addresses or strengthening authentication policies.

Importance of identifying failed logins:

- Detect brute-force and password-spray attacks

- Identify unauthorized user access attempts
- Monitor misuse of privileged accounts
- Support incident response and forensic investigations

Step 1: Generate Failed Login Attempts

Attempt SSH login with a wrong password:

ssh kali@127.0.0.1

```
(kali@kali)-[~]
$ ssh kali@127.0.0.1

kali@127.0.0.1's password:
Linux kali 6.18.5+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.5-1kali1 (2026-01-19) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Feb  3 10:49:59 2026 from 127.0.0.1
(kali@kali)-[~]
```

Step 2: Identify Failed Logins Using journalctl

View all failed authentication attempts: sudo

journalctl | grep -i failed

```
(kali@kali)-[~]
$ sudo journalctl | grep -i failed
[sudo] password for kali:
Nov 26 23:23:37 kali virtualbox-guest-utils[850]: 04:23:37.871824 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed
shared object file: No such file or directory
Nov 26 23:23:37 kali kernel: 04:23:37.871824 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:41 kali wireplumber[964]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: Failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:46 kali kernel: 04:23:46.780207 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:46 kali kernel: 04:23:46.796125 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:46 kali kernel: 04:23:46.821116 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:46 kali kernel: 04:23:46.840007 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:46 kali kernel: 04:23:46.852306 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:47 kali kernel: 04:23:47.376312 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.s
Nov 26 23:23:50 kali colord[1358]: failed to get edid data: EDID length is too small
Nov 26 23:23:51 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Av
Nov 26 23:23:51 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Av
Nov 26 23:23:53 kali dbus-daemon[805]: [system] Activation via systemd failed for unit 'dbus-org.bluez.service': Unit dbus-org.bluez.service not found.
Nov 26 23:23:53 kali dbus-daemon[1067]: [session uid=1000 pid=1067 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': U
e not found.
Nov 26 23:23:53 kali dbus-daemon[1067]: [session uid=1000 pid=1067 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': U
```

View SSH-specific failed logins: sudo

journalctl -u ssh | grep -i failed

```
(kali@kali)-[~]
$ sudo journalctl -u ssh | grep -i failed
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcloginUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
(kali@kali)-[~]
```

View authentication failure messages: sudo

journalctl | grep -i "authentication failure"

```
(kali@kali)-[~]
$ sudo journalctl | grep -i "authentication failure"
Nov 29 09:09:17 kali sudo[1621]: pam_unix(sudo:auth): authentication failure; logname=kali uid=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=kali
Jan 31 11:44:56 kali xfce4-screensaver-dialog[37078]: pam_unix(xfce4-screensaver:auth): authentication failure; logname=kali uid=1000 euid=1000 tty=0 ruser= rhost= user=kali
Jan 31 11:45:28 kali sudo[37409]: pam_unix(sudo:auth): authentication failure; logname=kali uid=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=kali
Jan 31 19:38:57 kali sudo[2228]: pam_unix(sudo:auth): authentication failure; logname=kali uid=1000 euid=0 tty=/dev/pts/0 ruser=kali rhost= user=kali
Feb 02 07:36:03 kali lightdm[1180]: pam_unix(lightdm:auth): authentication failure; logname= uid=0 euid=0 tty=0 ruser= rhost= user=kali
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=kali
(kali@kali)-[~]
```

Step 3: Analyze Failed Login Details From

the log output, observe:

- **Username** used for login
- **Source IP address**
- **Timestamp** of failed attempt
- **Service (SSH)**

4. Detect anomalies.

Anomalies in system and authentication logs are unusual or abnormal activities that deviate from normal user behavior or system usage patterns. Detecting anomalies helps identify security incidents, such as brute-force attacks, compromised accounts, insider threats, or unauthorized access. Log-based anomaly detection is a key part of security monitoring and digital forensics.

Common authentication-related anomalies:

- Multiple failed logins in a short time (brute-force attack)
- Login attempts at unusual times (e.g., late night)
- Logins from unknown or suspicious IP addresses
- Repeated attempts for non-existent users
- Sudden root or admin logins

Step 1: View Authentication Logs `sudo`

`journalctl -u ssh`

```
(kali㉿kali)-[~]
$ sudo journalctl -u ssh
Feb 03 10:45:42 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
Feb 03 10:46:46 kali sshd-session[3835]: Connection closed by 127.0.0.1 port 49538 [preauth]
Feb 03 10:47:43 kali sshd-session[4072]: Connection closed by 127.0.0.1 port 33820 [preauth]
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM
Feb 03 10:49:12 kali sshd-session[5036]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'kali')
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

Step 2: Detect Multiple Failed Logins (Brute Force)

`sudo journalctl | grep -i failed`

```

Feb 03 10:42:24 kali dbus-daemon[1300]: [session uid=1000 pid=1300 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': Unit evolution-source-registry.servic
e not found.
Feb 03 10:42:24 kali dbus-daemon[1300]: [session uid=1000 pid=1300 pidfd=5] Activation via systemd failed for unit 'evolution-source-registry.service': Unit evolution-source-registry.servic
e not found.
Feb 03 10:42:54 kali kernel: 15:42:54.781179 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:42:54 kali kernel: 15:42:54.781650 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Feb 03 10:49:12 kali sshd-session[5036]: failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:56:56 kali kernel: 15:56:56.324542 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:56:56 kali kernel: 15:56:56.324980 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:01 kali kernel: 15:57:01.578787 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:01 kali kernel: 15:57:01.583725 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:01 kali kernel: 15:57:01.591566 SHCLX11 Shared Clipboard: Converting X11 format index 0xa to VBox format 0x2 failed, rc=VERR_SHCLPB_NO_DATA
Feb 03 10:57:05 kali kernel: 15:57:05.685704 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:05 kali kernel: 15:57:05.687896 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:05 kali kernel: 15:57:05.689439 SHCLX11 Shared Clipboard: Converting X11 format index 0xa to VBox format 0x2 failed, rc=VERR_SHCLPB_NO_DATA
Feb 03 10:57:10 kali kernel: 15:57:10.845237 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:10 kali kernel: 15:57:10.846290 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 10:57:10 kali kernel: 15:57:10.848613 SHCLX11 Shared Clipboard: Converting X11 format index 0xa to VBox format 0x2 failed, rc=VERR_SHCLPB_NO_DATA
Feb 03 11:03:37 kali kernel: 16:03:37.749686 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 11:03:37 kali kernel: 16:03:37.757589 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 11:05:40 kali kernel: 16:05:40.380805 SHCLX11 Shared Clipboard: Reading clipboard data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
Feb 03 11:05:40 kali kernel: 16:05:40.381651 SHCLX11 Shared Clipboard: Requesting data in format 0x2 from host failed with VERR_SHCLPB_NO_DATA
--(kali@kali)-[~]

```

Step 3: Detect Login Attempts for Invalid Users sudo
journalctl | grep -i "invalid user"

Step 4: Detect Logins at Unusual Times sudo
journalctl | grep -i accepted

```

--(kali@kali)-[~]
$ sudo journalctl | grep -i accepted
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2

```

Step 5: Detect Suspicious IP Addresses sudo
journalctl -u ssh | grep -E "Failed|Accepted"

```

--(kali@kali)-[~]
$ sudo journalctl -u ssh | grep -E "Failed|Accepted"
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2

```

Step 6: Detect Privileged (Root) Access sudo
journalctl | grep -i root

```

--(kali@kali)-[~]
$ sudo journalctl | grep -i root
Nov 26 23:23:30 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38+kali-amd64 root=UUID=af89d218-8d5b-4054-a5b7-db050b5f62a7 ro quiet splash
Nov 26 23:23:30 kali kernel: Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38+kali-amd64 root=UUID=af89d218-8d5b-4054-a5b7-db050b5f62a7 ro quiet splash
Nov 26 23:23:30 kali kernel: ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-ff])
Nov 26 23:23:30 kali kernel: pci_bus 0000:00: root bus resource [io 0x0000-0x0cf7 window]
Nov 26 23:23:30 kali kernel: pci_bus 0000:00: root bus resource [io 0x0000-0xffff window]
Nov 26 23:23:30 kali kernel: pci_bus 0000:00: root bus resource [mem 0x000a0000-0x000bffff window]
Nov 26 23:23:30 kali kernel: pci_bus 0000:00: root bus resource [mem 0x80000000-0xfdf9ffff window]
Nov 26 23:23:30 kali kernel: pci_bus 0000:00: root bus resource [bus 00-ff]
Nov 26 23:23:30 kali kernel: Trying to unpack rootfs image as initramfs...
Nov 26 23:23:30 kali systemd[1]: systemd-fsck-root.service - File System Check on Root Device was skipped because of an unmet condition check (ConditionPathExists=!
Nov 26 23:23:30 kali systemd[1]: Starting systemd-remount-fs.service - Remount Root and Kernel File Systems ...
Nov 26 23:23:30 kali systemd[1]: Finished systemd-remount-fs.service - Remount Root and Kernel File Systems.
Nov 26 23:23:41 kali rtkit-daemon[969]: Successfully called chroot.
Nov 26 23:23:46 kali kernel: 04:23:46.859507 dnd Proxy window=0xa00001 (debug mode: false), root window=0x233 ...
Nov 26 23:25:18 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38+kali-amd64 root=UUID=af89d218-8d5b-4054-a5b7-db050b5f62a7 ro quiet splash
Nov 26 23:25:18 kali kernel: Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38+kali-amd64 root=UUID=af89d218-8d5b-4054-a5b7-db050b5f62a7 ro quiet splash
Nov 26 23:25:18 kali kernel: ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-ff])
Nov 26 23:25:18 kali kernel: pci_bus 0000:00: root bus resource [io 0x0000-0x0cf7 window]
Nov 26 23:25:18 kali kernel: pci_bus 0000:00: root bus resource [io 0x0000-0xffff window]
Nov 26 23:25:18 kali kernel: pci_bus 0000:00: root bus resource [mem 0x000a0000-0x000bffff window]
Nov 26 23:25:18 kali kernel: pci_bus 0000:00: root bus resource [mem 0x80000000-0xfdf9ffff window]
Nov 26 23:25:18 kali kernel: pci_bus 0000:00: root bus resource [bus 00-ff]
Nov 26 23:25:18 kali kernel: Trying to unpack rootfs image as initramfs...

```



```

Feb 03 11:09:01 kali CRON[14324]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 03 11:09:01 kali CRON[14326]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] 66 if [ ! -d /run/systemd/system ]; then /usr/lib/
Feb 03 11:09:01 kali CRON[14324]: pam_unix(cron:session): session closed for user root
Feb 03 11:09:03 kali sudo[14349]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -u ssh
Feb 03 11:09:03 kali sudo[14349]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:09:03 kali sudo[14349]: pam_unix(sudo:session): session closed for user root
Feb 03 11:10:14 kali sudo[14887]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 11:10:14 kali sudo[14887]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:10:14 kali sudo[14887]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:02 kali sudo[15762]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 11:12:02 kali sudo[15762]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:02 kali sudo[15762]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:17 kali sudo[15888]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 11:12:17 kali sudo[15888]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:17 kali sudo[15888]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:42 kali sudo[16085]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 11:12:42 kali sudo[16085]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:43 kali sudo[16085]: pam_unix(sudo:session): session closed for user root
Feb 03 11:13:25 kali sudo[16422]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -u ssh
Feb 03 11:13:25 kali sudo[16422]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:13:25 kali sudo[16422]: pam_unix(sudo:session): session closed for user root
Feb 03 11:14:08 kali sudo[16721]: _kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl

```

5. Correlate events.

Event correlation is the process of linking multiple log events from the same or different sources to understand the full sequence of an incident. Instead of viewing logs in isolation, correlation helps identify attack patterns, confirm security incidents, and determine cause–effect relationships. In Linux, correlating authentication, system, and network logs is essential for incident response and digital forensics.

Why event correlation is important:

- Reveals attack timelines (before, during, after)
- Confirms brute-force and compromise attempts
- Links failed logins to successful access
- Identifies attacker IPs and affected accounts
- Reduces false positives in security analysis

Step 1: Identify Failed Login Events sudo

`journalctl -u ssh | grep -i failed`

```

(kali@kali)-[~]
$ sudo journalctl -u ssh | grep -i failed
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcLoginUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2

```

Step 2: Identify Successful Login Events sudo

`journalctl -u ssh | grep -i accepted`

```

(kali@kali)-[~]
$ sudo journalctl -u ssh | grep -i accepted
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2

```

Step 3: Correlate Failed → Successful Logins (Same IP/User) sudo

`journalctl -u ssh | grep -E "Failed|Accepted"`


```
(kali@kali)-[~]
$ sudo journalctl -u ssh | grep -E "Failed|Accepted"

Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 11:03:11 kali sshd-session[11530]: Accepted password for kali from 127.0.0.1 port 49432 ssh2
```

Step 4: Correlate Authentication with User Session sudo

journalctl | grep -i "session opened"

```
(kali@kali)-[~]
$ sudo journalctl | grep -i "session opened"

Nov 26 23:23:40 kali lightdm[930]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=111) by (uid=0)
Nov 26 23:23:40 kali (systemd)[936]: pam_unix(systemd-user:session): session opened for user lightdm(uid=111) by lightdm(uid=0)
Nov 26 23:23:41 kali lightdm[930]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=111) by (uid=0)
Nov 26 23:23:45 kali lightdm[1034]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)
Nov 26 23:23:46 kali (systemd)[1045]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)
Nov 26 23:25:22 kali lightdm[707]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=111) by (uid=0)
Nov 26 23:25:22 kali (systemd)[714]: pam_unix(systemd-user:session): session opened for user lightdm(uid=111) by lightdm(uid=0)
Nov 26 23:25:22 kali lightdm[707]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=111) by (uid=0)
Nov 26 23:25:28 kali lightdm[810]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)
Nov 26 23:25:28 kali (systemd)[821]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)
Nov 26 23:35:01 kali CRON[6779]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Nov 26 23:39:01 kali CRON[8928]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Nov 26 23:45:01 kali CRON[12018]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)

Feb 03 10:59:18 kali sudo[9797]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:03:11 kali sshd-session[11530]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
Feb 03 11:04:30 kali sudo[12148]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:05:01 kali CRON[12404]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 03 11:05:29 kali sudo[12627]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:06:04 kali sudo[12901]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:09:01 kali CRON[14324]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 03 11:09:03 kali sudo[14349]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:10:14 kali sudo[14887]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:02 kali sudo[15762]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:17 kali sudo[15888]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:12:42 kali sudo[16085]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:13:25 kali sudo[16422]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:14:08 kali sudo[16731]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:15:01 kali CRON[17146]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 03 11:16:16 kali sudo[17691]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:17:02 kali CRON[18047]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 03 11:18:06 kali sudo[18553]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:18:43 kali sudo[18842]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:19:24 kali sudo[19174]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 03 11:20:18 kali sudo[19576]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
```

sudo journalctl | grep -i "session closed"

```
(kali@kali)-[~]
$ sudo journalctl | grep -i "session closed"

Nov 26 23:23:45 kali lightdm[930]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 26 23:23:45 kali lightdm[930]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 26 23:25:28 kali lightdm[707]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 26 23:25:28 kali lightdm[707]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 26 23:35:01 kali CRON[6779]: pam_unix(cron:session): session closed for user root
Nov 26 23:39:01 kali CRON[8928]: pam_unix(cron:session): session closed for user root
Nov 26 23:45:01 kali CRON[12018]: pam_unix(cron:session): session closed for user root
Nov 27 08:38:16 kali lightdm[804]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 27 08:38:16 kali lightdm[804]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 27 08:38:27 kali (sd-pam)[813]: pam_unix(systemd-user:session): session closed for user lightdm
Nov 27 08:39:01 kali CRON[2131]: pam_unix(cron:session): session closed for user root
Nov 27 08:45:01 kali CRON[5060]: pam_unix(cron:session): session closed for user root
Nov 27 08:55:01 kali CRON[9940]: pam_unix(cron:session): session closed for user root
Nov 29 09:04:51 kali lightdm[789]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 29 09:04:51 kali lightdm[789]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 29 09:05:01 kali (sd-pam)[798]: pam_unix(systemd-user:session): session closed for user lightdm
Nov 29 09:05:01 kali CRON[1570]: pam_unix(cron:session): session closed for user root
Nov 29 09:08:57 kali lightdm[705]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 29 09:08:57 kali lightdm[705]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Nov 29 09:09:01 kali CRON[1290]: pam_unix(cron:session): session closed for user root
Nov 29 09:09:07 kali (sd-pam)[714]: pam_unix(systemd-user:session): session closed for user lightdm
Nov 29 09:09:25 kali sudo[1621]: pam_unix(sudo:session): session closed for user root
Nov 29 09:09:36 kali sudo[1838]: pam_unix(sudo:session): session closed for user root
Nov 29 09:09:51 kali sudo[19551]: pam_unix(sudo:session): session closed for user root
```



```

Feb 03 10:43:51 kali sudo[1978]: pam_unix(sudo:session): session closed for user root
Feb 03 10:43:48 kali sudo[2416]: pam_unix(sudo:session): session closed for user root
Feb 03 10:43:58 kali sudo[2647]: pam_unix(sudo:session): session closed for user root
Feb 03 10:44:48 kali sudo[2758]: pam_unix(sudo:session): session closed for user root
Feb 03 10:44:53 kali sudo[3119]: pam_unix(sudo:session): session closed for user root
Feb 03 10:45:01 kali CRON[3197]: pam_unix(cron:session): session closed for user root
Feb 03 10:45:29 kali sudo[3388]: pam_unix(sudo:session): session closed for user root
Feb 03 10:45:42 kali sudo[3493]: pam_unix(sudo:session): session closed for user root
Feb 03 10:48:41 kali sudo[4913]: pam_unix(sudo:session): session closed for user root
Feb 03 10:54:24 kali sshd-session[5036]: pam_unix(sshd:session): session closed for user kali
Feb 03 10:54:33 kali sudo[7622]: pam_unix(sudo:session): session closed for user root
Feb 03 10:55:01 kali CRON[7844]: pam_unix(cron:session): session closed for user root
Feb 03 10:56:23 kali sudo[8460]: pam_unix(sudo:session): session closed for user root
Feb 03 10:59:18 kali sudo[9797]: pam_unix(sudo:session): session closed for user root
Feb 03 11:04:30 kali sudo[12148]: pam_unix(sudo:session): session closed for user root
Feb 03 11:05:01 kali CRON[12404]: pam_unix(cron:session): session closed for user root
Feb 03 11:05:29 kali sudo[12627]: pam_unix(sudo:session): session closed for user root
Feb 03 11:06:05 kali sudo[12901]: pam_unix(sudo:session): session closed for user root
Feb 03 11:09:01 kali CRON[14324]: pam_unix(cron:session): session closed for user root
Feb 03 11:09:03 kali sudo[14349]: pam_unix(sudo:session): session closed for user root
Feb 03 11:10:14 kali sudo[14887]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:02 kali sudo[15762]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:17 kali sudo[15888]: pam_unix(sudo:session): session closed for user root
Feb 03 11:12:43 kali sudo[16085]: pam_unix(sudo:session): session closed for user root
Feb 03 11:13:25 kali sudo[16422]: pam_unix(sudo:session): session closed for user root
Feb 03 11:14:08 kali sudo[16731]: pam_unix(sudo:session): session closed for user root
Feb 03 11:15:01 kali CRON[17146]: pam_unix(cron:session): session closed for user root

```

Step 5: Correlate Login Time with User Activity

last

```

(kali㉿kali)-[~]
$ last

kali      pts/1      127.0.0.1   Tue Feb 3 11:03 - still logged in
kali      pts/1      127.0.0.1   Tue Feb 3 10:49 - 10:54 (00:04)
kali      tty7      :0         Tue Feb 3 10:42 - still logged in
lightdm   tty7      :0         Tue Feb 3 10:42 - 10:42 (00:00)
kali      tty7      :0         Mon Feb 2 09:02 - 10:28 (01:26)
lightdm   tty7      :0         Mon Feb 2 09:01 - 09:02 (00:00)
kali      tty7      :0         Mon Feb 2 08:43 - 08:59 (00:15)
lightdm   tty7      :0         Mon Feb 2 08:43 - 08:43 (00:00)
kali      tty7      :0         Mon Feb 2 07:36 - still logged in
lightdm   tty7      :0         Mon Feb 2 07:35 - 07:36 (00:00)
kali      tty7      :0         Mon Feb 2 07:24 - 07:32 (00:08)
lightdm   tty7      :0         Mon Feb 2 07:23 - 07:24 (00:00)
kali      tty7      :0         Mon Feb 2 07:14 - still logged in
lightdm   tty7      :0         Mon Feb 2 07:14 - 07:14 (00:00)
kali      tty7      :0         Sat Jan 31 23:05 - 00:13 (01:07)
lightdm   tty7      :0         Sat Jan 31 23:05 - 23:05 (00:00)
kali      tty7      :0         Sat Jan 31 19:38 - 20:12 (00:33)
lightdm   tty7      :0         Sat Jan 31 19:38 - 19:38 (00:00)
kali      tty7      :0         Sat Jan 31 12:21 - still logged in
lightdm   tty7      :0         Sat Jan 31 12:21 - 12:21 (00:00)
kali      tty7      :0         Sat Jan 31 10:33 - 12:07 (01:34)
lightdm   tty7      :0         Sat Jan 31 10:33 - 10:33 (00:00)
postgres pts/2      127.0.0.1   Sat Jan 31 09:29 - 09:29 (00:00)
kali      pts/2      127.0.0.1   Sat Jan 31 09:24 - still logged in
postgres pts/2      127.0.0.1   Sat Jan 31 09:23 - 09:23 (00:00)
kali      tty7      :0         Sat Jan 31 08:35 - still logged in

```

6. Learn SIEM Basics

SIEM (Security Information and Event Management) is a security solution that collects, centralizes, and analyzes logs from multiple sources such as servers, firewalls, applications, and network devices. SIEM helps security teams detect threats, correlate events, generate alerts, and respond to incidents in real time.

Key functions of SIEM:

- Log collection and centralization
 - Event correlation from multiple sources
 - Real-time threat detection
 - Alert generation and incident reporting
 - Compliance and audit support
- Examples of SIEM tools:**

Splunk, ELK Stack (Elasticsearch, Logstash, Kibana), IBM QRadar, ArcSight

7. Write Alerts

Security alerts are notifications generated when suspicious or malicious activity is detected based on predefined rules or thresholds. Alerts help security analysts respond quickly to potential threats and minimize damage.

Characteristics of good security alerts:

- Clear and specific (what, when, where)
 - Triggered by defined conditions
 - Prioritized based on severity
 - Actionable (suggest response steps)
- Examples of alert conditions:**
- More than 5 failed login attempts in 1 minute
 - Successful login after multiple failures
 - Root login from unknown IP
 - Login outside normal working hours

Alerts are usually generated by SIEM systems after correlating multiple log events.

8. Document Findings

- Analyzed different types of system logs to understand system and user activities
- Authentication logs revealed successful and failed login attempts
- Multiple failed login attempts indicated possible unauthorized access attempts
- Abnormal login patterns were identified as security anomalies
- Correlation of failed and successful logins helped reconstruct event timelines
- Observed that repeated failures followed by success may indicate brute-force attacks

- Learned the importance of SIEM for centralized log monitoring and analysis
- Defined alert conditions for suspicious activities such as repeated failed logins
- All observations and analysis were properly documented for future reference
- Documentation supports incident response, audits, and security improvement efforts